

EXPERIMENT 6

Student Name: Jay Varshney

UID: 24BDA70206

Branch: AIT-CSE

Section/Group: 24AIT_KRG-G2

Semester: 5

Subject Name: ADBMS

Subject Code: 24CSP-333

Experiment 6 -- To study and implement advanced SQL queries using Joins, Correlated Query's and Conditional statement.

Aim : To study and implement correlated subqueries in SQL by retrieving employees whose salaries are among the top three highest distinct salary values within their respective departments, while correctly handling duplicate salary values using aggregate functions and joins

Question :6 (i) Calculate Apple and Orange Daily Sale difference

SQL Query:

```
SELECT sale_date, SUM(CASE WHEN fruit = 'apples' THEN sold_num ELSE -sold_num END) as s
FROM Sales
```

GROUP BY sale_date

	sale_date date	s bigint
1	2020-05-04	-1
2	2020-05-02	0
3	2020-05-03	20
4	2020-05-01	2

EXPERIMENT 6

Question: 6 (ii) Determine the type of Node

SQL Query:

```
SELECT DISTINCT t.id, (  
CASE
```

```
    WHEN t.p_id is NULL THEN 'Root'
```

```
    WHEN l.id is NULL THEN 'Leaf'
```

```
    ELSE 'Inner'
```

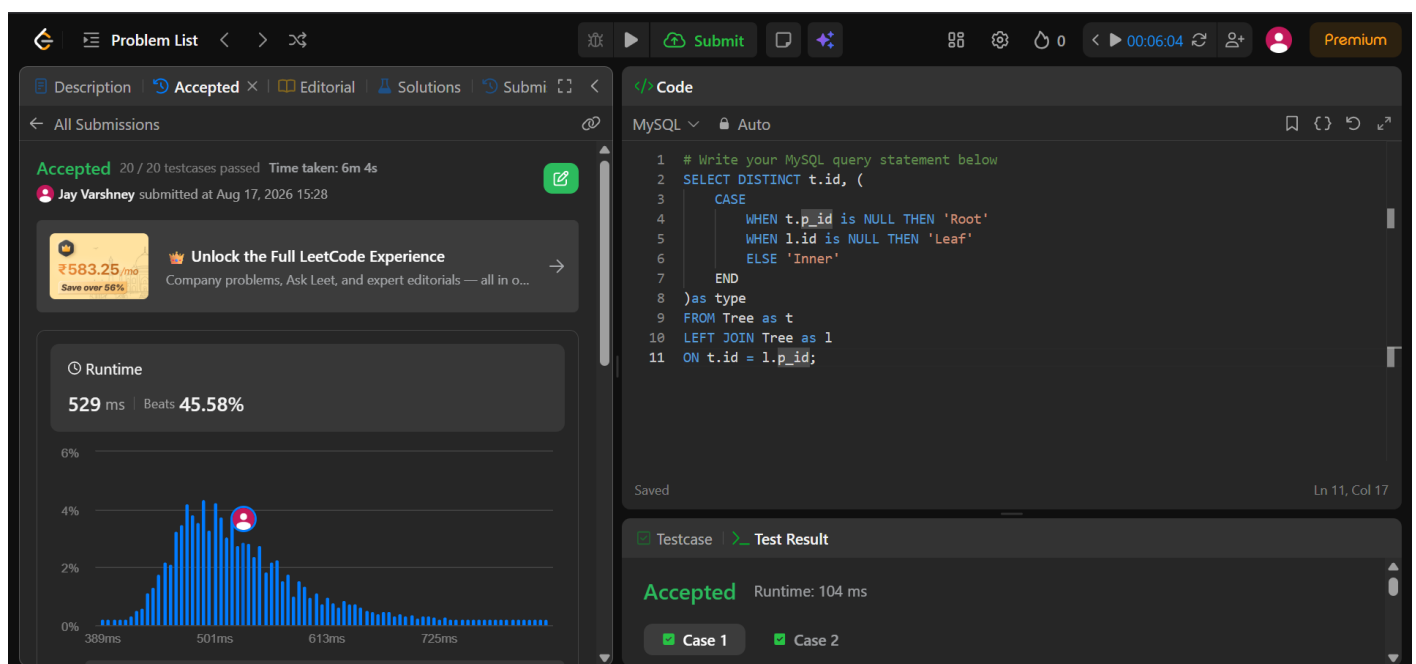
```
END
```

```
)as type
```

```
FROM Tree as t
```

```
LEFT JOIN Tree as l
```

```
ON t.id = l.p_id
```



The screenshot shows a LeetCode submission interface. On the left, the 'All Submissions' tab is active, showing a submission by 'Jay Varshney' that is 'Accepted' with 20/20 testcases passed and a runtime of 6m 4s. Below this, a 'Runtime' section shows a performance of 529 ms, which beats 45.58% of other submissions. A histogram shows the distribution of runtimes, with the user's submission marked. On the right, the 'Code' editor shows the SQL query:

```
1 # Write your MySQL query statement below
2 SELECT DISTINCT t.id, (
3     CASE
4         WHEN t.p_id is NULL THEN 'Root'
5         WHEN l.id is NULL THEN 'Leaf'
6         ELSE 'Inner'
7     END
8 )as type
9 FROM Tree as t
10 LEFT JOIN Tree as l
11 ON t.id = l.p_id;
```

 The bottom of the interface shows the 'Test Result' section, which is 'Accepted' with a runtime of 104 ms, and both 'Case 1' and 'Case 2' are marked as passed.

EXPERIMENT 6

Learning Outcome

After completing this experiment, the learner will be able to:

1. **Understand and implement correlated subqueries** to solve SQL problems involving row-wise comparisons.
2. **Apply aggregate functions** such as COUNT() and DISTINCT to retrieve top-N records within groups.
3. **Use joins and filtering techniques** to combine data from multiple tables and generate meaningful reports.
4. **Analyze and solve complex SQL queries** involving grouped data, ranking logic, and handling duplicate values effectively.